

ADVANCED OPTICAL MATERIALS

Supporting Information

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Structure Classes Using Generative Deep Learning

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Global Inverse Design Across Multiple Photonic Structure Classes Using Generative Deep Learning: Supporting Information

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Training Dataset

The training dataset for deep learning consists of precisely 18,770 metasurface unit cell designs (12,632 MIM and 6,138 hybrid dielectric structures). These designs were derived from seven starting shape templates: cross, square, ellipse, bow-tie, H, V, and tripole-shaped. As shown in Figure S1a, parameter sweeps were performed on each shape (for the MIM structures) to produce geometric variations. The tabulated parameter sweeps are captured within $3.2 \times 3.2 \mu\text{m}^2$ unit cells. For the hybrid dielectric structures, the same parameter sweeps were scaled by $\times 2.34 \mu\text{m}$, and the unit cell dimensions for this group of structures were $7.5 \times 7.5 \mu\text{m}^2$. Both sets of structures are represented as $64 \times 64 \times 3$ pixel images. Figure S1b shows several image pairs, which illustrate examples of finalized metasurface designs from each shape template. Images on the left are the color-encoded images used for deep learning, and images on the right represent the corresponding 3D physical models. As described in the main text, the colors on the image are used to indicate the metasurface class, resonator geometry, material choice, and dielectric thickness. Specifically, each structure possesses a 100 nm thickness gold substrate. For each MIM structure, the metal resonator is a 100 nm layer of gold, silver, or aluminum, while the dielectric material is Al_2O_3 with a thickness of 100 nm, 200 nm, or 300 nm. For each hybrid dielectric structure, the dielectric resonator is zinc selenide, silicon, or germanium, and its thickness is 500 nm, 750 nm, or 950 nm.

Full-wave simulations were performed on each structure (under p-polarization at normal incidence) to produce a corresponding 800-point absorption spectrum across the mid-infrared wavelength range. Low quality designs which exhibited ‘flat’ (maximum absorption is less than 0.2) or ‘noisy’ (mean-squared error, or MSE, between the spectra and its average is greater than 0.05) spectral responses were removed from the training set to maximize the model’s utility and performance. Figure S2 illustrates the peak absorptions and resonance wavelengths of all the spectra training dataset, organized by full width at half maximum (FWHM). The distribution of absorption spectra reveals that the MIM structures ($\text{FWHM} \geq 0.4 \mu\text{m}$) cover a wide range of peak amplitudes and resonance wavelengths from 4-10 μm , while a majority of the hybrid

dielectric structures (FWHM $\leq 0.2 \mu\text{m}$) exhibit responses from 7.5-8.5 μm . The range of responses in the training dataset may be extended in future studies to enhance the network's predictive capabilities. On a distributed high-performance computing cluster with four dedicated compute nodes per simulation, where a node has a minimum of four 64-bit Intel Xeon or AMD Opteron CPU cores and 8 GB memory, each FDTD simulation took approximately 5 minutes to complete. Therefore, our training dataset equates to approximately 65 days of simulation time.

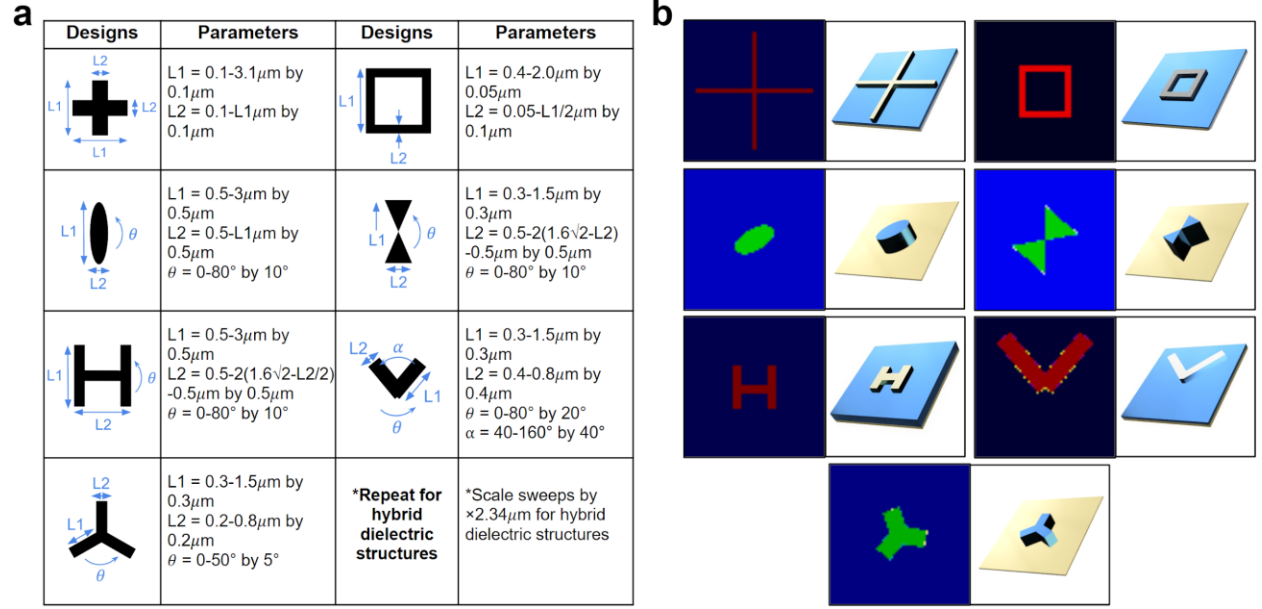


Figure S1. (a) 2D images of template shapes used to derive unit cell designs for both classes of metasurfaces. The range of variation allowed for each parameter is listed to the right of the associated shape. (b) Color-encoded 2D images (left) representing metasurface resonators. Images with red colors represent MIM structures while images with green colors represent hybrid dielectric structures. Shades of blue represent the dielectric thickness for both classes of structures. 3D models of each resonator are shown to the right of their 2D representation.

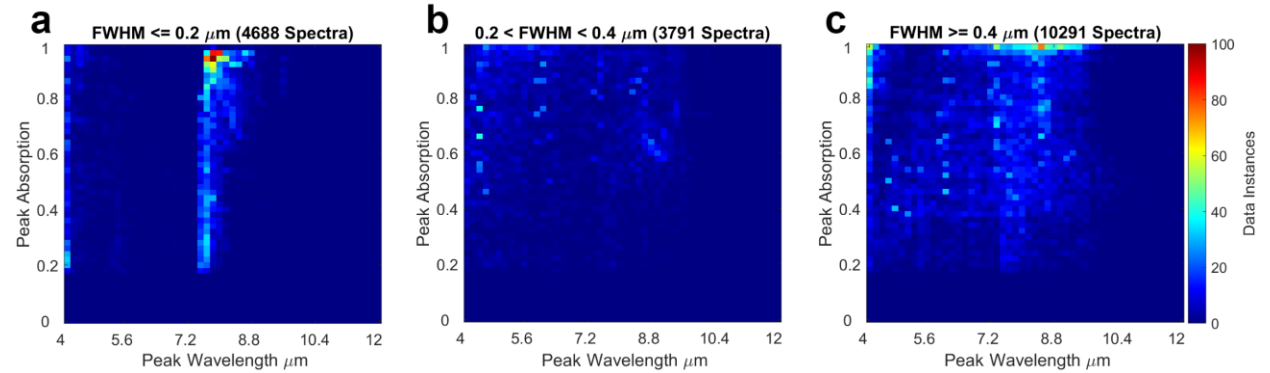


Figure S2. Visualization of peak absorption and wavelength values in the training set as distribution heatmaps. Regions of higher red-intensity indicate increased data instances within this range. The individual spectra within the training set are sorted into three subsets based on the FWHM of their absorption peak: (a) FWHM less than or equal to $0.2 \mu\text{m}$ (b) between $0.2 \mu\text{m}$ and $0.4 \mu\text{m}$, and (c) greater than $0.4 \mu\text{m}$.

Network Architecture Design and Optimization

Implemented in the PyTorch framework, the cDCGAN consists of two networks: a generator and a discriminator (illustrated in Figure S3). The optimized generator contains five transposed convolutional layers (with 1200, 1024, 512, 256, and 128 input channels or feature maps), while the discriminator has five convolutional layers (with 6, 64, 128, 256, and 512 input channels or feature maps). Each transposed convolutional layer in the generator is followed by a batch normalization and ReLU (rectified linear unit) activation layer, instead of the final layer, where a Tanh (hyperbolic tangent) activation is used. Similarly, in the discriminator, each convolutional layer is followed by a batch normalization and Leaky ReLU layer, and the final layer possesses a Sigmoid activation. At the generator input layer, the 800-point absorption spectra are concatenated with 400-point latent vectors to yield 1200-point input vectors. For the discriminator input, the absorption spectra are passed through a fully-connected layer and reshaped into a $64 \times 64 \times 3$ matrix. These matrices were then concatenated with the real and generated images to form $64 \times 64 \times 6$ inputs for the discriminator. Model training was performed on an NVIDIA Titan RTX GPU and took approximately 30 minutes to complete.

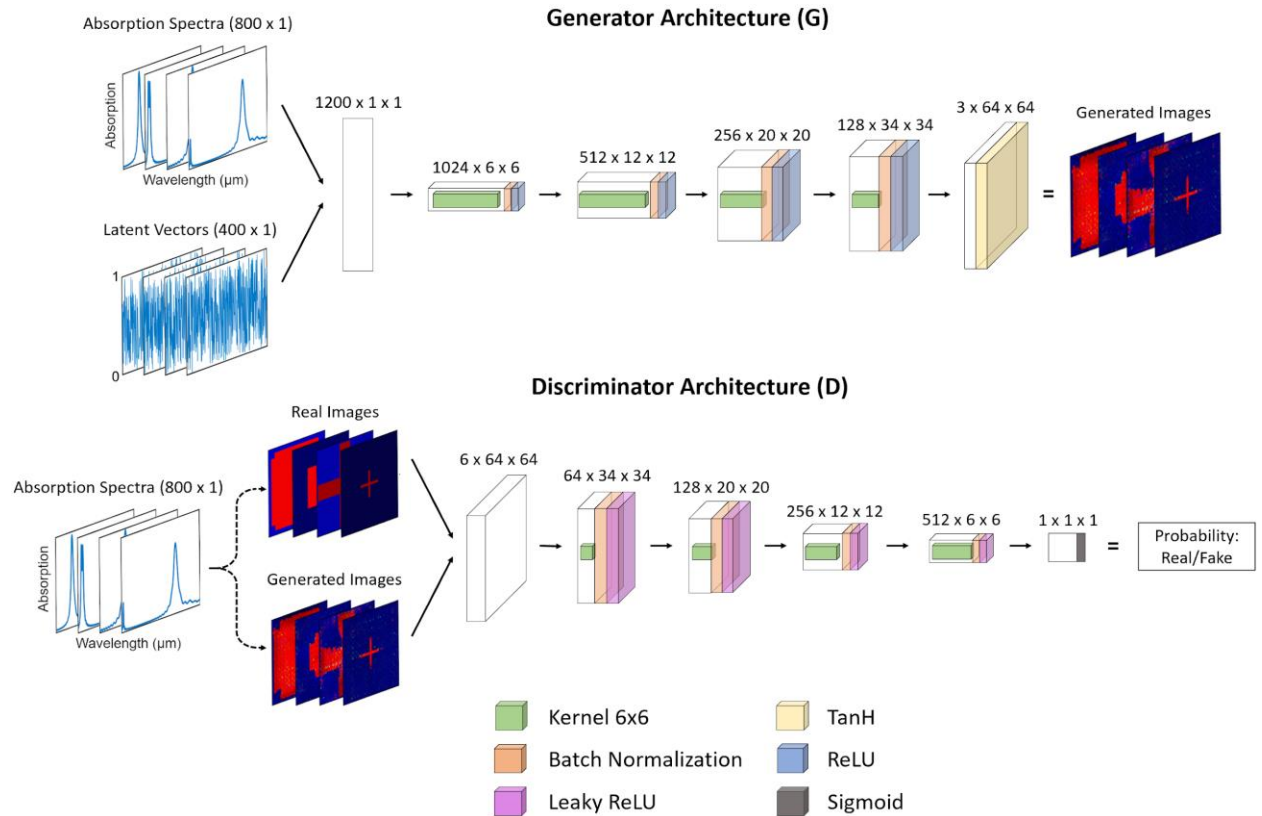


Figure S3. Schematic of the generator and discriminator architectures implemented in our cDCGAN model. Input and output types are shown for each layer along with the layer types and dimensions.

Evaluation of Multiparametric Encoding Methods

We tested the efficacy and performance of three different material and structural parameter encoding methods. Figure S4 shows the training progression of each encoding method at various epochs. As seen in Figure S4a, the first encoding method uses several neurons to represent the parameters by embedding them into a single row and column of pixels within the topological image (for each parameter). Specifically, in this encoding scheme, an initial 64×64 pixel image is converted to a 66×66 image, where the new rows and columns represent the material property and dielectric thickness of the metasurface design. The second method (Figure S4b), similar to the first, encodes the material and structure parameters as two rows and columns per parameter (yielding a 68×68 image). The third and final method that was investigated (Figure S4c) involves encoding the parameters into discrete ‘RGB’ channels of colored images, as described in the main text.

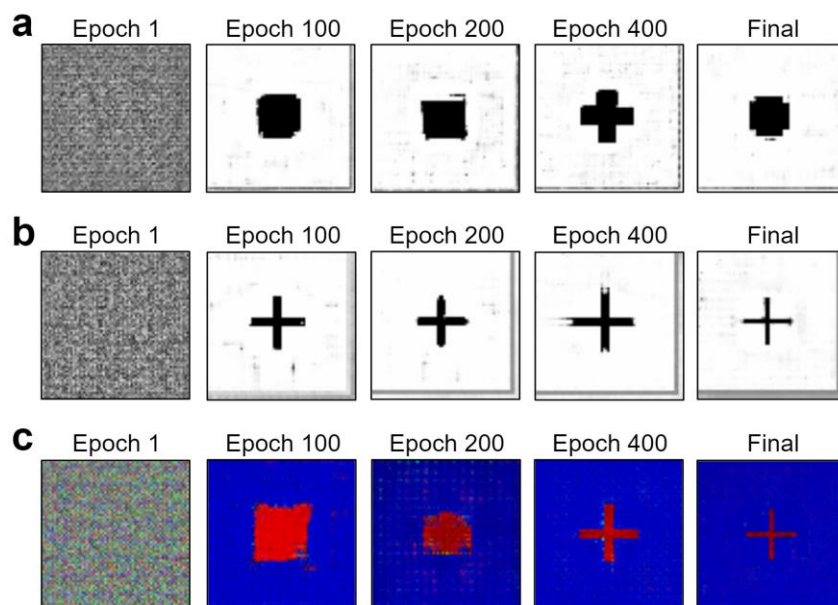


Figure S4. Examples of the generative model’s training progress for three implementations of multiparametric encoding over hundreds of epochs. The tested implementations embedded parameter information as (a) a single-row and column vector concatenated with the 2D image, (b) double-row and column vectors concatenated with the image, and (c) normalized values within the ‘RGB’ channels of colored images.

Each encoding method was applied to the training dataset, and the corresponding datasets were separately used to train the cDCGAN. Hyperparameter tuning was performed via grid search, and the results for each dataset are presented in Table S1. Here, various feature maps, kernels, batch sizes, epochs, and miscellaneous pre- and post-processing steps were tested. Listed feature map sizes represent the lowest denomination of maps used in the intermediate layers

(between the first and last layers) of the generator and discriminator. Reported losses are derived from the validation dataset.

Table S1. Hyperparameter optimization for various multiparametric encoding methods.

Highlighted cells indicate the lowest validation loss for the corresponding method.

Model	Feature Maps (G/D)	Kernel Size	Batch Size	Misc.	Epochs	Validation Loss (MSE)
Single Row/Column Encoding (MIM Only)						
1	32/16	4	128		750	0.0264
2	64/32	4	128		750	0.0215
3	64/32	4	128		1000	0.025
4	66/66	4	16	Gaussian Filter ($\sigma=0.75$)	500	0.0269
5	66/66	4	16	Gaussian Filter ($\sigma=0.75$)	750	0.0332
6	66/66	4	128	Gaussian Filter ($\sigma=0.75$)	750	0.0127
7	66/66	4	128		750	0.0135
8	66/66	6	16	Gaussian Filter ($\sigma=0.75$)	500	0.0351
9	66/66	6	16	Gaussian Filter ($\sigma=0.75$)	750	0.0405
Double Row/Column Encoding (MIM Only)						
10	68/68	4	128		750	0.0413
11	68/68	5	16		250	0.0379
12	68/68	5	16		500	0.0292
13	68/68	5	32		500	0.0291
14	68/68	5	64		500	0.0157
15	68/68	6	16		500	0.0128
16	68/68	6	16	Gaussian Filter ($\sigma=1$)	500	0.0168
17	68/68	6	16	Gaussian Filter ($\sigma=0.75$)	500	0.0112
18	68/68	6	32		500	0.0177
19	68/68	6	32	Gaussian Filter (1)	500	0.0183
20	68/68	6	32	Normal Distribution (z)	500	0.0281
21	68/68	6	32	Noise 350 pts (z)	500	0.0328
22	68/68	6	32	Noise 450 pts (z)	500	0.0294
23	68/68	6	68		500	0.0247
24	68/68	6	128		500	0.0211
Color-Encoding (MIM Only)						
25	64/32	6	16	Boundary Thresh. (0.2) + GF	750	0.0044
26	64/32	6	16	Boundary Thresh. (0.1) + GF	750	0.0153
27	64/32	6	32	Boundary Thresh. (0.2) + GF	500	0.013
28	64/32	6	32	Boundary Thresh. (0.1) + GF	500	0.0142
29	64/32	6	32	Boundary Thresh. (0.2) + GF	750	0.0111
30	64/32	6	32	Boundary Thresh. (0.1) + GF	750	0.0116
31	64/32	6	64	Boundary Thresh. (0.2) + GF	500	0.0179
32	64/32	6	64	Boundary Thresh. (0.2) + GF	750	0.0285

We note that the highest-performing hyperparameters for a specific encoding method was not typically the optimal model for other encoding methods. As a result, each encoding method was optimized independently of prior models. To expedite our training efforts, only the MIM structures were used for the first round of optimization. Across all the explored encoding methods, we observe that the color-encoding approach yielded the lowest validation loss

(0.0044) and highest performance. In addition to hyperparameter tuning, a Gaussian filter (GF) with binary thresholding offered substantial performance gains (where σ is the standard deviation of the Gaussian kernel) and significant reduction in noise-related artifacts, while modifying the latent vector (z) size and distribution (uniform to normal) resulted in no noticeable improvements.

When training the cDCGAN with the color-encoded images, the discriminator frequently overpowered the generator and resulted in mode collapse. Thus, we reduced the size of the discriminator’s layers in comparison to the generator to balance the two networks. Furthermore, we developed an image processing workflow to convert the generated images into full 3D metasurface designs. Here, each generated image is decoded into three components: a resonator-only image, a material property value, and a dielectric thickness value. The resonator image specifies the existence (black pixels) or absence (white pixels) of planar features. These pixels are obtained by determining the boundaries between major color gradients on the GAN-generated color images (*e.g.*, red-blue or green-blue transition points), thus a boundary conversion threshold was applied in order to find the exact transition points. Here, we determined that the optimum threshold was a fifth of the maximum resonator color intensity (shown as 0.2 in Table S1 and S2) in the red or green color channel. If a pixel position possessed a red or green pixel value that exceeded the threshold, then the existence of a physical structure was indicated here. Prior to determining these feature boundaries, a binary classification is performed by calculating the dominant class-specific color, which is used to classify the structure (if red pixels are greater than green, then the structure is MIM, and vice versa for hybrid dielectric). The purpose of this procedure is to filter any stray red pixels that may be intermingled with green and vice versa, and to assign the appropriate boundary conditions and unit cell dimensions to the FDTD model. As described in the main text, material property and thickness values are then calculated by taking the average pixel-values in their respective channels (based on structure classification), then reversing the normalization performed in the encoding step.

Table S2. Final hyperparameter optimization with the entire training dataset. The highlighted cell indicates the model with the lowest validation loss.

Model	Feature Maps (G/D)	Kernel Size	Batch Size	Misc.	Epochs	Validation Loss (MSE)
Color-Encoding (MIM + DM Only)						
33	64/32	6	16	Boundary Thresh. (0.2) + GF	750	0.0128
34	64/32	6	16	Boundary Thresh. (0.2) + GF	1000	0.0094
35	64/32	6	32	Boundary Thresh. (0.2) + GF	500	0.0136
36	64/32	6	32	Boundary Thresh. (0.2) + GF	750	0.0086
37	64/32	6	32	Boundary Thresh. (0.2) + GF	1000	0.0135
38	64/32	6	32	Boundary Thresh. (0.2) + GF	700	0.0115
39	64/32	6	32	Boundary Thresh. (0.2) + GF	800	0.0125
40	128/64	6	16	Boundary Thresh. (0.2) + GF	500	0.0076
41	128/64	6	16	Boundary Thresh. (0.2) + GF	750	0.0106

42	128/64	6	16	Boundary Thresh. (0.2) + GF	1000	0.0125
43	128/64	6	32	Boundary Thresh. (0.2) + GF	1000	0.0144

After identifying that the color-encoding strategy resulted in the highest design accuracy (or lowest validation loss), a second round of optimization was conducted on the entire training dataset. As shown in Table S2, with the optimized post-processing procedure determined in the previous section, the highest-performance model was trained with a kernel size of 6, batch size of 16, 128 base generator feature maps, 64 base discriminator feature maps, and for 500 epochs.

Batch Testing

We evaluated the cDCGAN’s performance across a wide range of new inputs by creating 200 Fano-shaped and Lorentzian-shaped spectra with amplitudes ranging from 0.5-0.9, and resonance wavelengths ranging from 5-9 μm , for 400 total test spectra. Figure S5 illustrates a comparison of 50 individual responses within this ‘batch’ test. Each tiled plot is presented with 4-12 μm wavelength and 0-1 absorption axes limits. Here, we observe that the Fano-shaped responses (Figure S5a) are most accurate between resonance wavelengths of 7.5-8.5 μm , while the Lorentzian-shaped responses (Figure S5b) maintain strong matches across the entire test data range. Regions of low accuracy correspond to the areas that are not well-represented by the training dataset (shown in Figure S2). Therefore, the performance of the cDCGAN may be improved by expanding the training data and design space.

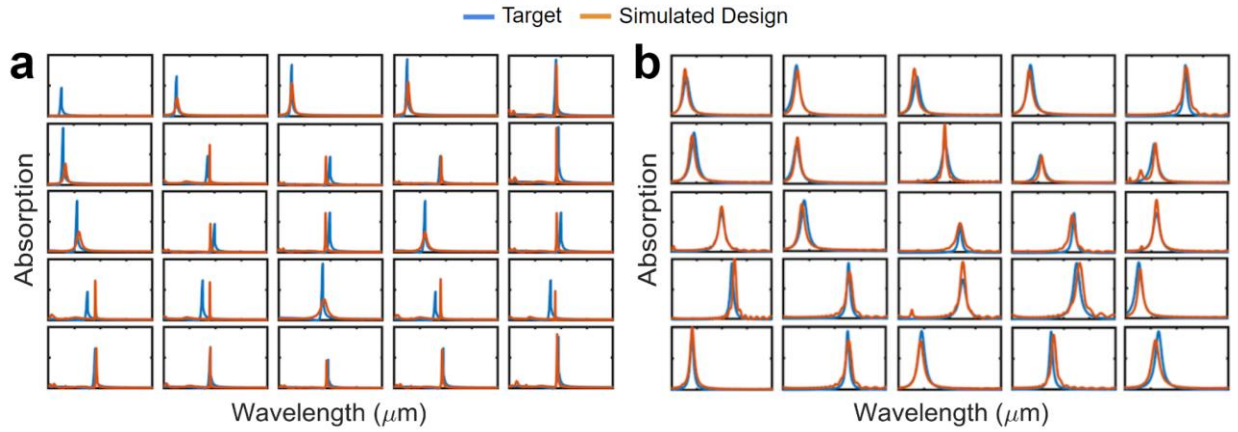


Figure S5. Test results of the cDCGAN using a diverse range of newly-constructed (a) Fano-shaped and (b) Lorentzian-shaped inputs. Blue lines represent input targets while the orange lines represent simulated designs produced by the cDCGAN.

Designing Complex Alloyed Structures

Though we limited this study to the application of uniform materials, we note that the devised color-encoding strategy is capable of representing spatially-varying material properties along the entire physical structure. This in turn sets the stage for future studies with much greater design complexity, such as 3D or complex metal alloy-based structures, with potentially greater control over the electromagnetic spectrum. A demonstration of this capability is shown in Figure S6, where the different shades of color on the cDCGAN-generated MIM structures are converted into different metals (shown in the inset images) based on their individual plasma frequencies, rather than the average over the channel. Notably, the simulated alloyed structures yield similar responses to the uniform material structures, beyond which there are no distinguishable advantages in the particular design space that was explored. Therefore, future studies utilizing a wider range of dissimilar materials (as well as the application of fabrication constraints tailored towards alloy-based design) may produce device properties that extend beyond the uniform material domain to gradient-index materials.

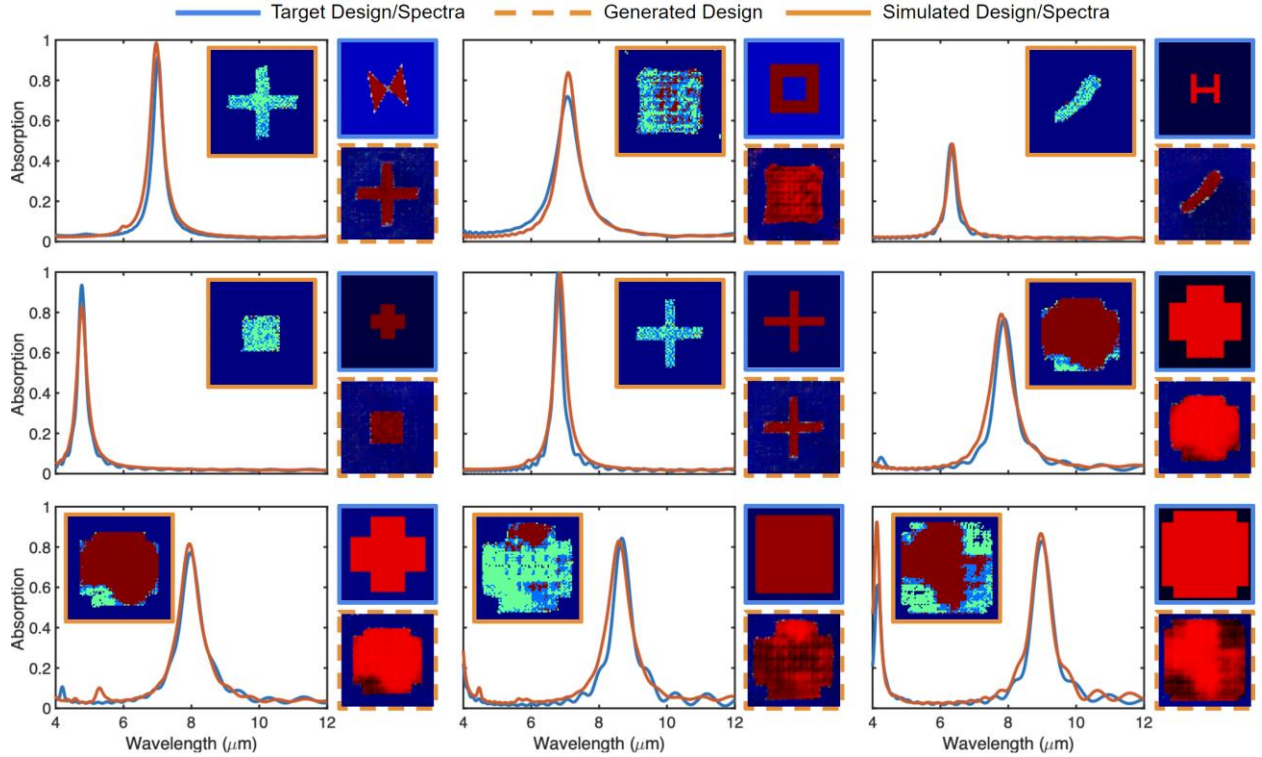


Figure S6. Demonstration of metal alloy-based structure design using the color-encoded cDCGAN. Blue lines represent the input spectra and reference designs. Dashed-orange lines represent the cDCGAN output, and solid-orange lines are alloyed structures created using the color gradients. We note that the simulated results match well with the input targets. The fabricability of these structures could potentially be improved with the addition of fabrication constraints such as minimum feature size.

Latent Vector Sampling and Model Validation

Since the GAN may produce a distribution of designs with potentially varying degrees of accuracy, ten different latent vectors were generated for each target spectrum, which were then used as inputs to the network. Each design is verified using numerical simulation, then the design (and corresponding latent vector) with the lowest mean-squared error to the target is used as the final design. Figure S7 shows the distribution of designs across different latent vectors for several input targets (a Lorentzian function centered at 7.2 μm and at 7.8 μm), where we observe that all the generated designs have over 90% accuracy in comparison to the input target. Following this procedure, Figure S8 presents a series of tests performed with inputs that originate from the training dataset. An equivalent analysis for the validation dataset can be found in the main text.

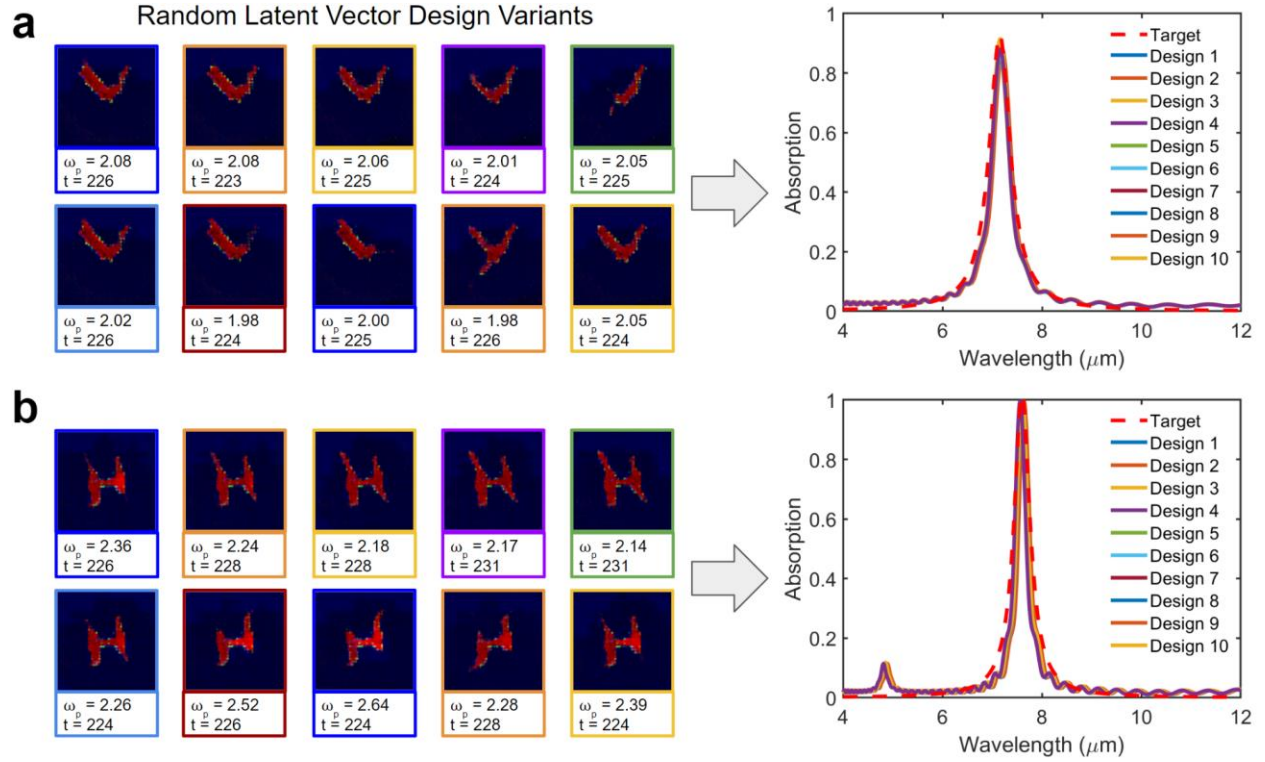


Figure S7. GAN-produced design variants achieved by pairing the target spectrum with 10 randomly-generated latent vectors. Input targets for Lorentzian functions centered at (a) 7.2 μm and (b) 7.8 μm are indicated by the dashed red lines. For each target, the design with the lowest mean-squared error is used as the final design.

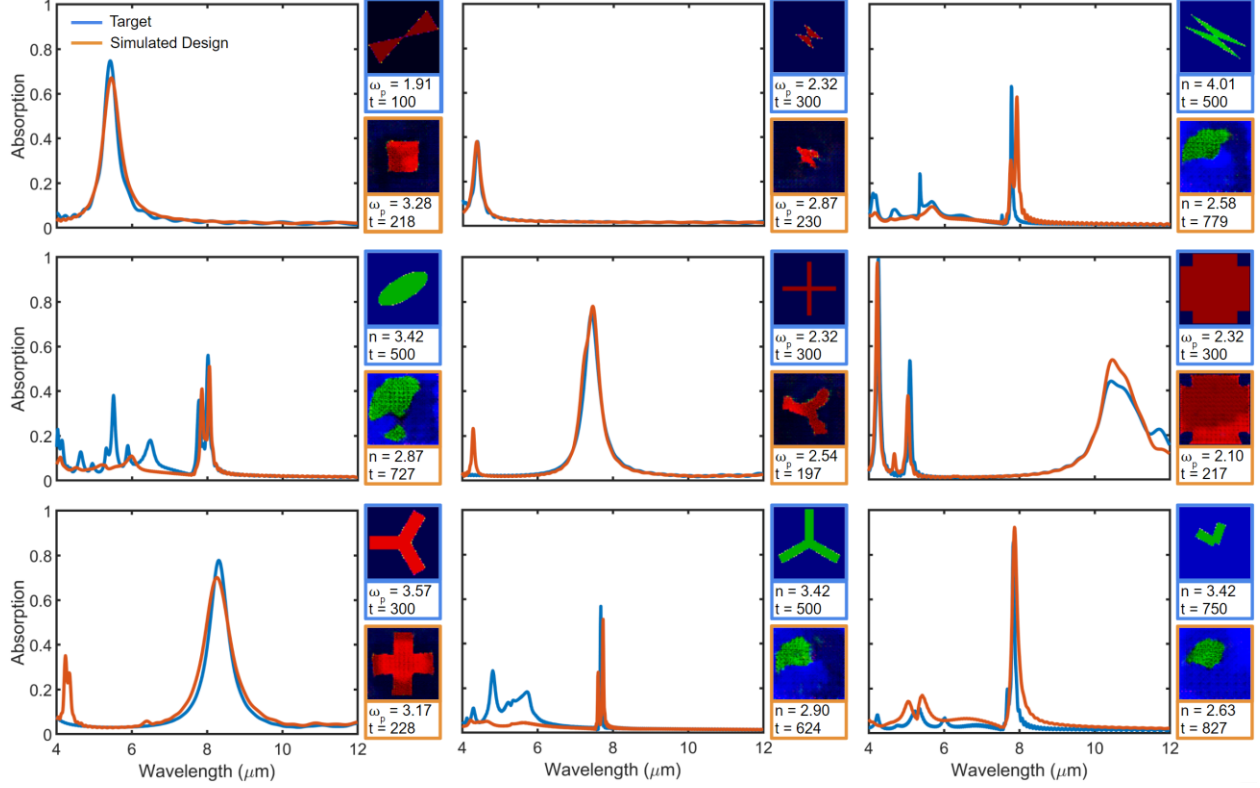


Figure S8. Randomly selected absorption spectra from the training dataset (in blue) which were designated as input targets for the cDCGAN. The simulated spectra of the cDCGAN-synthesized designs (in orange) are plotted alongside the targets for comparison. Images representing the respective structures are shown to the right of each plot, with material and thickness information below each image. Units for plasma frequency (ω_p) values are in PHz and thicknesses (t) are in nanometers. The results here reveal that the network is not copying the training dataset, but to a degree, it is identifying the underlying relationships between structure, material, metasurface class, and optical response to provide new yet accurate solutions that extend beyond the training dataset.